

MSc Large Databases

Module outline: from a single query to systems that scale

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1 • Aims

How databases stay correct and fast as the data grows

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What students can do by the end

- Design a schema and reason about normalisation and trade-offs
- Read a query plan and explain why one runs and another scales
- Reason about transactions, concurrency, and recovery under failure
- Choose between relational, NoSQL, and warehouse architectures for a given workload

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The arc of the module

- 1. The relational model and advanced SQL:** schema design, normalisation, windowing, CTEs, views
- 2. Storage and indexing:** the memory hierarchy, pages, B+ trees, hash indexes
- 3. Query processing and optimisation:** join algorithms, cost-based plans, statistics
- 4. Transactions and concurrency:** ACID, isolation levels, locking, multi-version concurrency
- 5. Recovery and reliability:** write-ahead logging, checkpoints, durability under failure
- 6. Scale and distribution:** partitioning, replication, CAP, NoSQL families, warehousing and OLAP, big-data engines

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6 units

one module, six units: from the relational model to distributed, large-scale systems

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Tools and datasets

- **PostgreSQL** and **MySQL** for relational work; a document store (**MongoDB**) and a columnar/warehouse engine for the scale unit
- Query-plan inspection as a habit: read what the optimiser actually does, not what you assume
- Production-scale datasets, so indexing and optimisation decisions have visible consequences

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How it is taught

- Lectures connect each layer of the stack to the failure it prevents
- Hands-on labs on query optimisation and transaction behaviour, then a design-and-build task on a large dataset
- The recurring question: what happens to this design when the data is a hundred times larger?

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Why it matters

Every analytics and AI system rests on the data layer. Understanding the internals is what separates a query that runs from one that scales.

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In summary

Core texts: Silberschatz, Korth & Sudarshan, Database System Concepts · Ramakrishnan & Gehrke, Database Management Systems · Kleppmann, Designing Data-Intensive Applications
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